ARTICLE IN PRESS

Sports Medicine and Health Science xxx (xxxx) xxx

Contents lists available at ScienceDirect

Sports Medicine and Health Science

journal homepage: www.keaipublishing.com/smhs

Does wet-cupping therapy improve repeated sprint ability, perceived wellness, and rating of perceived exertion in young active males?

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ARTICLE INFO

Keywords:
Alternative medicine
Complementary medicine
Hijama
Short-term maximal performance
Sports medicine

ABSTRACT

Wet-cupping therapy (WCT) is one of the oldest known medical techniques, used as a traditional and complementary therapy with a wide application all around the world for general health. Research on the effects of WCT on sports performance are sparse and inconsistent. Thus, we aimed to explore the effects of WCT on repeated sprint ability, wellness, and exertion in young active males. Forty-nine active adult males (age: [28 ± 5] years; body height [177 ± 8] cm; body mass: [79 ± 7] kg; body mass index: [25.4 ± 1.8] kg/m^2) were selected for the study. The participants performed a running-based sprint test on two separate occasions (Control and Post-WCT). WCT was performed 24 h before the testing session. They completed the Hooper questionnaire to assess their well-being (***i.e.***, sleep, stress, fatigue, and soreness) before each session. The rating of perceived exertion (RPE) was collected after each testing session. A higher maximum power ($p < 0.05$, effect size [*ES*] = 0.6), mean power ($p < 0.01$, *ES* = 0.5) and minimum power ($p < 0.01$, *ES* = 0.6) were recorded post-WCT as compared to Control session along with a better perceived sleep ($p < 0.01$, *ES* = 0.85). Perceived stress ($p < 0.01$, *ES* = 0.6) and RPE ($p < 0.001$; *ES* = 1.1) were lower during the post-WCT compared to the Control session. The present findings demonstrated that WCT moderately enhanced repeated sprint ability and had positive effects on perceived sleep, stress, and exertion. WCT may be an effective ergogenic aid to improve repeated sprint ability and general well-

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https://doi.org/10.1016/j.smhs.2023.09.007
Received 4 February 2023; Received in revised form 7 September 2023; Accepted 15 September 2023
Available online xxxx
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Please cite this article as: Dergaa I et al., Does wet-cupping therapy improve repeated sprint ability, perceived wellness, and rating of perceived exertion in young active males?, Sports Medicine and Health Science, https://doi.org/10.1016/j.smhs.2023.09.007

being in young adult males. Future large-scale multicentric clinical studies are paramount to confirm the results of our study.

Abbreviations list

CT	Cupping therapy
DOMS	delayed onset muscle soreness
ES	Effect size
NO	Nitric oxide
RAST	Running-based anaerobic sprint test
RPE	Rating of perceived exertion
WCT	Wet cupping therapy

1. Introduction

Cupping therapy (CT), an ancient technique of alternative and complementary medicine, has been used as a treatment for several conditions such as herpes zoster, vitiligo, arterial hypertension, rheumatoid arthritis, and migraine headaches.[1,2] The two popular techniques of CT are 'Dry' and 'Wet' CT.[2,3] Dry CT consists of sucking the skin into a cup without any incision. Whereas, in wet-CT (WCT), the skin is scarified to draw blood and serum into the cup.[2] WCT is the most used and effective technique of CT.[2] Recently, the interest in cupping has re-appeared and consequently, numerous studies have started to investigate the potential benefits of this technique.[2–6] During the 2016 Rio-de-Janeiro Olympic Games in Brazil, many athletes were observed with CT-induced large red circles all over their bodies.[7]

Despite the long history and recent renewal of interest in the use of WCT, little is known about its benefits, especially in sports performance.[7] A small number of studies about WCT in sports have focused on pain release,[2,8] and injury prevention,[8] but little is known about the effect of WCT on physical performance.[7] A recent study suggested that WCT might have beneficial effects on the central nervous system (*e.g.*, higher frontal cortex activation), thereby may enhance the muscle response during physical activity.[2] Besides, WCT may contribute to an increase of endogenous substances such as nitric oxide (NO), which induce beneficial biological changes.[2] NO causes arteries vasodilatation[9] and contributes to a reduction in vascular resistance[9] and blood pressure,[9] and genesis inhibition of platelet aggregation and adhesion.[2,9,10] WCT was reported to improve sleep quality in healthy individuals.[11,12] Better sleep quality has been shown to increase sprint performance,[13] reduce muscle damage, and enhance antioxidant defence during repeated sprint exercise.[14,15] Moreover, Sun et al.[16] reported that the perceived exertion during the exercise was lower in thirty adolescent gymnasts following CT compared to controls.

Repeated high-intensity sprint efforts require both anaerobic and aerobic metabolic pathways in energy production.[16] Because of its acceptance as a good procedure for assessment of anaerobic power and capacity, the running-based anaerobic sprint test (RAST) has been used as a standard procedure for anaerobic evaluation tests for different athletes populations[13,14,17–19] such as soccer,[20,21] and futsal players.[22] In their pilot study, Ekrami et al.[23] reported that Karate practitioners receiving WCT showed lower exercise-induced inflammatory markers as interleukin-6 and α-tumour necrosis factor and. This suggests that WCT could reduce the inflammatory response to vigorous exercise by enhancing muscle tissue integrity (*e.g.*, lower muscle damage) and/or increasing oxygen and nutrients transporter (*i.e.*, red blood cell family).[23]

Based on the existing literature, WCT could improve sleep quality (*e.g.*, inducing a more restorative sleep), increase blood flow (*i.e.*, increasing oxygen and nutrients availability to the active muscle), relieve pain (*i.e.*, enhancing the joy of exercising), and decrease the exercise-induced inflammatory markers (*i.e.*, reducing the muscle damage).[9] The abovementioned mechanisms would potentially enhance repeated sprint performance.[14,18,19]

This study aimed to explore the potential effects of WCT on repeated sprint performance, perceived wellness, and exertion in young active males. We hypothesized that WCT might have a positive effect on repeated sprint performance, perceived wellness, and exertion in young active males.

2. Methods

Ethical approval

The study was designed according to the guidelines of the Helsinki Declaration for conducting human experimentation and was approved by the ethical committee of the Farhat HACHED Hospital, Sousse, Tunisia (Approval N°: FH/20200411). Each participant was informed of the purpose, procedure, risks, and study details and consequently signed an informed consent form.

2.1. Participants

Forty-nine healthy moderately active adult males (age: [28 ± 5] years; height [177 ± 8] cm; body mass: [79 ± 7] kg; body mass index: [25.4 ± 1.8] kg/m^2) volunteered to participate in this study.

Participants were meticulously selected based on a rigorous screening process to ensure they did not have any orthopaedic, neuromuscular, skin, coagulopathy, and/or cardiovascular issues. All participants, as part of their mandatory regimen for police officers, engaged in resistance and endurance training approximately four times per week. This consistent training, combined with their yearly fitness assessments, indicates a high level of physical fitness among the participants. Such a regimen is designed to prepare them for the immediate reaction demands frequently placed on police officers in their professional roles. Police officers are often required to be on high alert, ready for potentially demanding and hazardous situations. These scenarios necessitate peak physical and mental performance, drawing parallels with the challenges faced by athletes in high-intensity sports. The nature of a police officer's duties, which involves physically challenging activities in potentially perilous situations while safeguarding the public and preventing crimes, further underscores the relevance of studying this demographic in the context of our research objectives.

Participants were instructed to avoid excessive training and exercise, and to consume caffeinated drinks or alcohol 48 h before the testing sessions. Additionally, they were instructed not to use any antioxidant or anti-inflammatory drugs, or consume foods rich in antioxidants or polyphenols (***e.g.***, Curcuma, dark chocolate, blueberries, red wine, grapes, tea, and cherries) during and for at least three weeks before the experimental period.[24–26] A familiarization session was conducted seven days before the first testing session to allow participants to become habituated to the protocol. Furthermore, a healthy nutritional intake and hydration strategy were provided simultaneously. A consecutive dietary record was completed over seven days to assess the adequacy of nutrient intake. Everyone's dietary composition was estimated using the food composition tables published by the Tunisian National Institute of Statistics in 1978 and the Bilnut 4 software package (SCDANutrisoft, Cerelles, France). Estimated daily nutriment data showed that total calorie, macronutrient, and micronutrient intakes were within reference ranges for healthy adults (data not shown).

2.2. Procedures

This was a single-arm; control (Control) and testing (Post-cupping) sessions' interventional study, to assess the effects of WCT on repeated sprint performance. The two trials, Control and Post-cupping, were conducted two weeks apart in a counter-balanced and randomized order. During the Control session, participants completed Hooper's questionnaire[27] to assess the participants' perceived wellness before the RAST.[28,29] Rating of perceived exertion (RPE) was collected 10 min after the RAST.[28]

During the Post-cupping trial, participants completed the WCT 24 h before the trial. They completed the same control tests as for the Control session in the same order (Fig. 1). All sessions were performed indoors, and at the same time of the day (*i.e.*, between 7:00 a.m. and 9:00 a.m.) to minimize the effects of diurnal variations in the measured variables.[30,31] Sessions were conducted three days away from the full moon days to avoid the effect of circa-lunar variation on the measured parameters.[13,32] The mean ± standard deviation (*SD*) of the ambient temperature and relative humidity were (23 ± 2)℃ and 62% ± 4%, respectively.

Special attention was given to the potential influence of occupational factors on the tolerance and efficacy of the WCT. To ensure consistency and minimize confounding variables, police officers were selected from a singular working shift. The WCT session was strategically scheduled for the evening of their subsequent day off. Furthermore, both the running test and the associated questionnaires were administered on their non-working day. This approach was adopted to ensure that the results were not inadvertently influenced by external occupational factors.

2.3. Wet cupping therapy

The WCT session was conducted by a certified and experienced health care practitioner, adhering to the standardized method.[33] Participants were positioned in a prone position during the therapy. Prior to the application of the cups, five points on the bilateral paraspinal areas of the neck and thoracic spine were sanitized using a specific antiseptic solution to prevent any potential infections.

The cupping pressure was meticulously calibrated. Disposable cups were positioned at the designated regions (Fig. 2). A moderate vacuum-induced cupping pressure (*i.e.*, 250–300 mmHg) was applied, ensuring a consistent and standardized pressure across all participants. This pressure was determined based on prior studies and was maintained to ensure both efficacy and safety. After a duration of 3 min, the cups were removed. Sterile lancets were employed at the sites where cupping pressure was exerted, scarifying to a depth of approximately 2 mm. This depth was chosen to ensure effective blood and serum drainage while minimizing potential discomfort and risk to the participants.

Subsequently, the vacuum cups were repositioned for a second time to facilitate the drainage of blood and serum.[33] Povidone-iodine was applied to the scarified sites to prevent potential infections and promote healing.[34] Upon completion of the WCT, sterile pads infused with an additional layer of the anti-infective agent were utilized to cover the scarified sites, offering an added layer of protection. To ensure the utmost safety during the procedure, an emergency doctor and nurse equipped with an emergency response kit were present in the application room. Their presence ensured immediate response to any unforeseen complications, although, fortunately, their intervention was not necessitated at any point during the procedure.

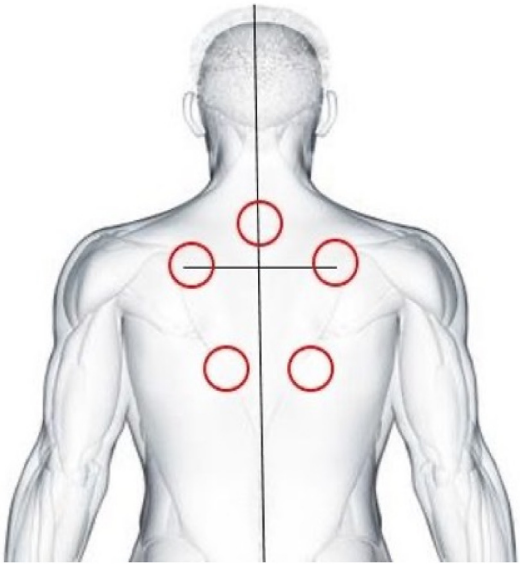

Fig. 1. Posterior torso showing points on the skin in which wet cupping therapy was applied.

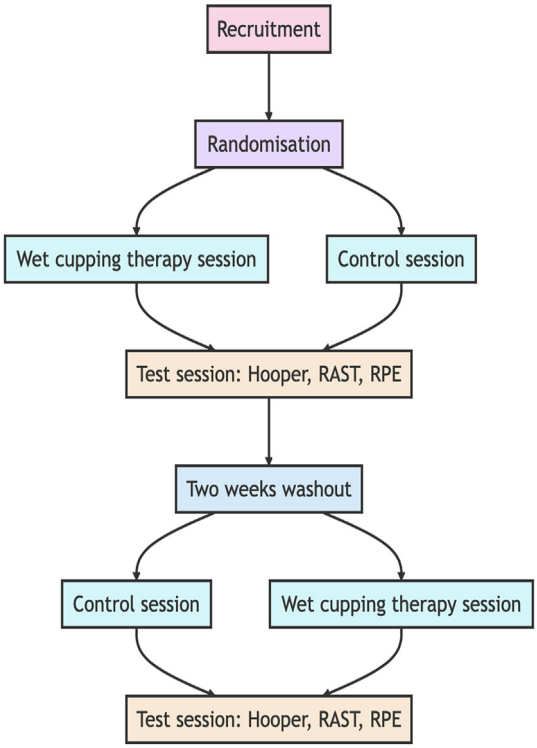

Fig. 2. Experimental design. **Hooper**: Hooper questionnaire. **RAST**: Running-based anaerobic sprint test. **RPE**: Rating of perceived exertion.

2.4. Running-based anaerobic sprint test

The repeated sprint ability was evaluated by the RAST. The test consisted of 6 x 35-m maximal sprints interspersed with 10-s passive recovery periods. The RAST is a valid and reliable test for assessing short-term power and capacity. The maximum power, mean power, and minimum power were calculated.[35,36]

2.5. Wellness monitoring

Participants were asked to respond subjectively about the **(i)** Quality of sleep during the preceding night, **(ii)** Quantity of perceived stress, **(iii)** Fatigue, and **(iv)** Muscle soreness, they felt at the moment of assessment (pre-testing sessions) according to Hooper questionnaires,[29] with lower subjective ratings always indicate better scores.

2.6. Exertion monitoring

The RPE was evaluated using the Foster scale (CR-10) to collect subjective estimations of RPE during the testing session.[28,37]

2.7. Data analysis

Variables distribution was assessed by the one-sample Kolmogorov-Smirnov test, and data were normally distributed. A two-tailed paired sample *t*-test was used to compare mean differences in the Control and Post-cupping results. Additionally, the effect size (*ES*) was calculated for pairwise comparison. The magnitude of ES was interpreted as trivial ($<$ 0.2), small (0.2–0.6), moderate (0.6–1.2), large (1.2–2.0), and very large ($\geq$ 2.0). All statistical analyses were conducted using the statistical package for the social sciences (SPSS, version 18.0, SPSS Inc, Chicago, IL, USA). The level of significance was set at $p < 0.05$.

3. Results

Differences between Control and Post-cupping sessions in repeated sprint ability, perceived wellness, and perceived exertion are detailed in Table 1.

WCT improved the rates of perceived sleep quality, stress quantity, and exertion with lower rates registered during the Post-cupping compared to the Control session ($p < 0.01$, $ES = 0.6$–1.1, mean difference = 0.46–0.97). WCT did not affect the perceived fatigue and the quantity of delayed onset muscle soreness (DOMS) measured before the RAST (Fig. 3; Table 1).

WCT improved repeated sprint ability with higher powers (***i.e.***, maximum, minimum and mean powers) registered during the Post-WCT compared to the Control session ($p < 0.001$, $ES = 0.5$–0.6, mean difference = 30.8–68.6 W) (Fig. 4; Table 1).

4. Discussion

The main findings of this study were that WCT moderately improved the perceived sleep quality during the next night and enhanced the repeated sprint ability in the morning of the next day while decreasing the pre-test perceived stress and post-effort RPE. This is a pioneer study to provide information on the effects of WCT on repeated sprints' physical performance and pre-test perceived wellness. Thus, our hypothesis is confirmed. However, future large-scale multicentric clinical studies are paramount to further confirm our results.

Our findings revealed moderate but significant improvements in maximum, minimum, and mean powers Post-cupping compared to Control. Several studies have examined the benefits of WCT on indirect markers of muscle damage.[3,23] Ekrami et al.[23] reported in a pilot study involving 21 male karate practitioners that inflammatory markers (*i.e.*, Interleukin-6 and α-tumour necrosis factor), were significantly lower in athletes who received WCT immediately post-exercise, suggesting such therapy may mitigate the inflammatory response to vigorous exercise among martial arts athletes. Furthermore, it has been reported that WCT may enhance the production of endogenous NO which acts as a vasodilator, antineoplastic and antimicrobial agent.[2] Thus, dilated blood vessels shall assure better nutrients and oxygen delivery to the working muscles during physical exercise and might contribute to consequently enhancing physical performance.[38] Unfortunately, NO concentrations were not measured in our study. Although speculative, the WCT-induced increase in NO may explain the improved repeated sprint ability in this study.

In another study, Abdullahi et al.[31] reported that WCT increased the frontal brain β- and δ-activities in electroencephalograph measures of 36 healthy males compared to control electroencephalograph recordings. The authors concluded that WCT might have beneficial effects on the central nervous system, by contributing to an increase in the secretion of neuro-humoral hormones (*e.g.*, B-endorphin and adrenocortical),[31] which could increase the muscle response during physical activity. This could partially explain the higher repeated sprint performance in the current study. We concluded that WCT might have beneficial effects on repeated sprint performance. In this regard, future large-scale multicentric clinical studies are needed to further investigate this.

Our findings revealed a better perception of sleep quality after WCT. These results are in line with the study of Cikar et al.[12] who reported an enhanced sleep quality (explored via the Pittsburgh Sleep Quality Index parameters) after WCT in healthy people. The authors explained this improvement by the cuppings' analgesic effects. In this context, it has been revealed that WCT induces a washout of pain-mediating substances,[39,40] and increases serum endogenous opioids.[12] Additionally, WCT could have excretory benefits.[2] Thus, through the exerted vacuum pressure, large macromolecules (*e.g.*, globulins; antibodies, and rheumatoid factor) can be removed along with the cupping blood and serum.[39] Furthermore, triglycerides, cholesterol, low-density lipoprotein, heavy metals, and oxidants have been reported to significantly decrease after WCT.[6,40] The association between the decrease of heavy metals and oxidants along with the analgesic effects of WCT can partially explain the better-perceived sleep quality during the WCT subsequent night compared to Control. Thus, our results indicated that WCT might improve sleep quality.

Perceived level of stress was lower 24 h Post-cupping. The lower perceived stress could also be due to the immediate analgesic effect generated by the WCT.[33] Furthermore, a lower level of perceived stress has been positively correlated to better sleep perception.[40] Thus, the improved level of sleep perception generated by WCT could be associated with the lower perceived stress observed in our study.

Our results revealed no effect on perceived fatigue and muscle soreness Post-cupping compared to Control. Despite lower stress and better sleep quality, it seems that the effects of WCT are not observable in participants with normal levels of fatigue and DOMS (Fig. 2). Indeed, our participants were instructed to avoid heavy physical exercise before the testing days and therefore, reported normal levels of 'rested DOMS and stress'. Further investigations might study the effect of WCT in participants with high levels of fatigue and DOMS (***c.f.*** the potential effects of physiological responses to WCT mentioned above).[33]

Despite the higher repeated sprint performance, we report a moderate decrease in the RPE Post-cupping compared to Controls. The latter finding could be explained by the analgesic effects of WCT, in line with earlier findings of some authors.[16] However, we highlight that the RPE may be influenced by sociological, psychological, and environmental factors of the participants.[31]

Although the fact that WCT is considered a safe technique of alternative and complementary medicine, in some cases, it could trigger some complications such as scar formation, bruises, infection, and abscesses,

Table 1
Mean differences between Control and Post-cupping test sessions in the rating of perceived wellness and exertion and the anaerobic sprint test performance.

Variables	Mean difference (mean value at Post-cupping – mean values at Control session)	95% confidence interval of mean difference	*t*	*p*	Effect size
Sleep	−0.97	0.59 to 1.36	5.15	0.001	0.9
Stress	−0.46	0.17 to 0.76	3.21	0.002	0.6
Muscle soreness	−0.06	−0.40 to 0.52	0.26	0.79	0.1
Fatigue	−0.26	−0.2 to 0.73	1.13	0.26	0.2
Rating of perceived exertion	−0.71	0.48 to 0.93	6.28	0.001	1.1
Maximum power (W)	68.64	−90.88 to −46.41	−6.2	0.001	0.6
Minimum power (W)	30.76	−42.78 to −18.74	−5.14	0.001	0.5
Mean power (W)	43.32	−52.08 to −34.57	−9.94	0.001	0.6

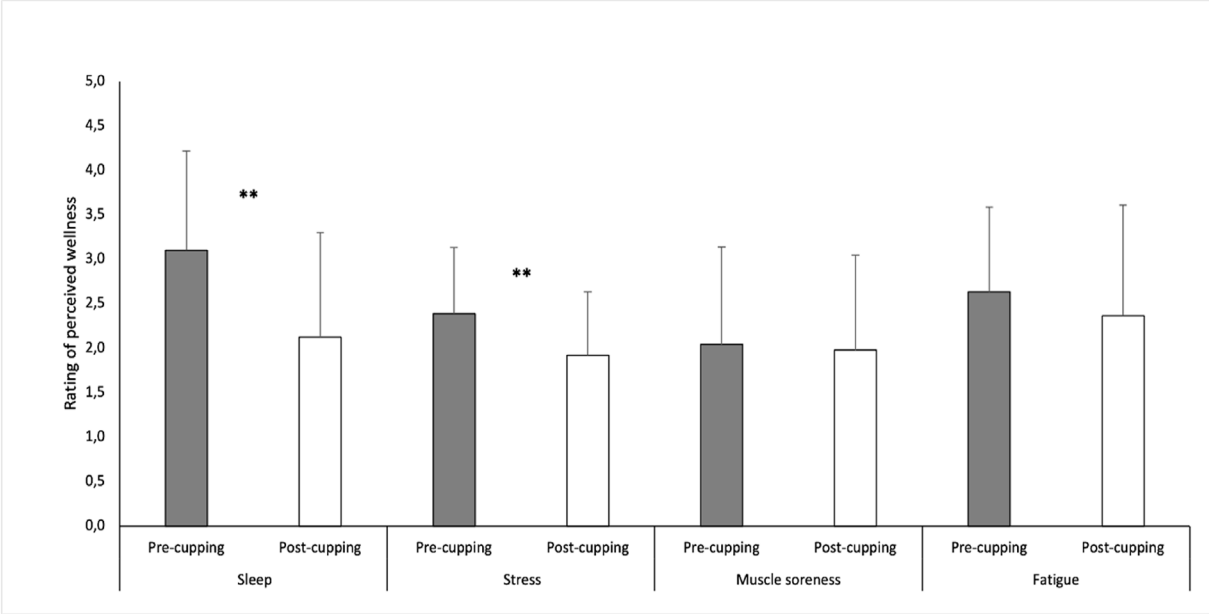

Fig. 3. Comparison between Control session (Pre-cupping) and Post-WCT session (Post-cupping) on rating of perceived wellness (measured before the RAST) and exertion (measured after the RAST).
RAST: Running based anaerobic sprint test. **: Significant difference between Control and Post cupping sessions ($p < 0.01$).

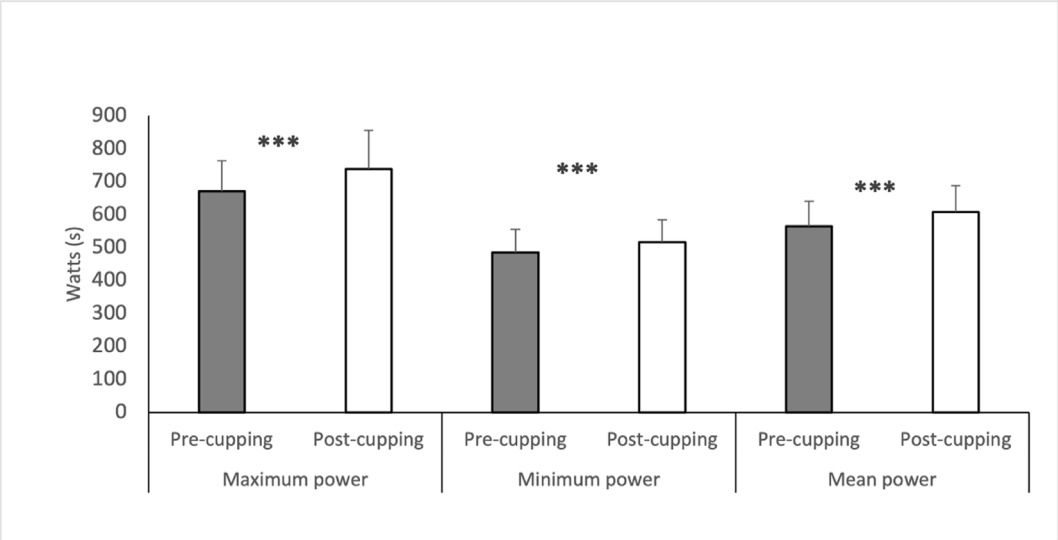

Fig. 4. Comparison between Control session (Pre-cupping) and Post-WCT session (Post-cupping) on the anaerobic sprint test performance. ***: Significant difference between Control and Post-cupping session ($p < 0.001$).

which are mostly the result of unqualified and inexperienced cupping therapists.[41] Thus, we recommend the WCT be performed by highly skilled and experienced therapists and qualified medical personnel, under strict aseptic precautions and infection control measures as per guidelines to avoid any complications.

Future studies should focus on the metabolic changes due to WCT to improve understanding of its potential modes of action. However, to the best of the authors' knowledge, the present investigation is the first study examining the effects of WCT on repeated sprint ability, perceived exertion, and wellness in a cohort study.

Our study presents some limitations. First, the effect of WCT on repeated sprint performance, perceived wellness, and exertion was assessed once; 24 h post-intervention, further research is warranted to investigate the time course of these effects. Second, the recruited participants were not athletes, and the study was not a part of a training program inducing fatigue and muscle soreness, which would explain the absence of a significant effect on fatigue and muscle soreness assessed before the test sessions. However, it would be more informative if muscle soreness was investigated 24 h and 48 h following the repeated sprint exercise rather than before. Future studies should evaluate WCT on muscle soreness in the athletic population (preferably during a training program, where muscle soreness is already high). The metabolic responses to WCT were not assessed. Further, sleep quality was only assessed by the perceived wellness questionnaire. Last, measuring some biological parameters (***e.g.***, NO and plasma lactate) would be helpful to understand the real mechanisms by which WCT influences physical performance. The causal relationship between WCT and physical performance remains unclear and needs to be further explored. Additional

studies are required to reach conclusive results on the efficiency, safety, durability, and real mechanisms of action of WCT through larger randomized controlled clinical trials, systematic reviews, and meta-analysis to assess its potentially beneficial effects.

5. Conclusions

Our findings revealed that WCT moderately enhanced repeated sprint performance and had positive effects on perceived sleep, stress, and exertion. Hence, WCT is potentially useable as a simple and effective ergogenic aid to help improve repeated sprint ability and general well-being in the young adult population.

Submission statement

All authors have read and agree with manuscript content. The manuscript has not been published and is not under consideration for publication elsewhere.

Ethical approval statement

The study was designed according to the guidelines of the Helsinki Declaration for conducting human experimentation and was approved by the ethical committee of the Farhat HACHED Hospital, Sousse, Tunisia (Approval N°: FH/20200411). Each participant was informed of the purpose, procedure, risks, and study details and consequently signed an informed consent form.

Authors' contributions

ID and HBS: Conception and design.

ID, HBS, MR, AS, AA, KC, BK: Analysis and interpretation of the data.

ID, HBS, AG, MR, AS, AA, RAF, MSF, KI, MT, TM, MAD, NS, OH, KW, KC, BK: Drafting of the paper.

ID, BSH, AG, MR, AS, AA, RAF, MSF, KI, MT, TM, MAD, NS, OH, KW, KC, BK: Revising it critically for intellectual content.

All authors gave their final approval to the version that will be published.

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors.

Availability of data and materials

The data related to this manuscript is available from the first author upon reasonable request.

Acknowledgement

We express our gratitude to ChatGPT-4 for its invaluable assistance in generating Fig. 2.

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